

Effects of anthropogenic disturbances on bird functional diversity: A global meta-analysis

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ARTICLE INFO

Keywords:

Avian
Biodiversity
Ecosystem functioning
Ecological traits
Human perturbations

ABSTRACT

Anthropogenic disturbances represent the main threat to biodiversity around the globe, yet effects are not restricted to species loss. Assessing the functional diversity, which measures the range and value of ecological traits of organisms, can provide a more direct link between biodiversity and ecosystem functioning. Therefore, as species' responses to disturbances depend on their ecological traits, ecosystem functions are likely to be also imperiled by human perturbations. Although the literature on the effects of disturbances on functional diversity has been increasing, results are highly varied. Here, we provided the first comprehensive global meta-analysis on the effects of human disturbances on functional diversity of birds. We also performed subgroup analyses by region, habitat specialization, functional trait, functional diversity index and disturbance type. Our results indicate a consistent negative effect of disturbances on bird functional diversity, regardless of the variability caused by the different predictor variables considered. Specifically, we revealed that studies that were conducted in the tropics, focused on feeding traits and calculated functional diversity and richness indices presented negative responses to disturbances. Additionally, the functional diversity of birds, regardless of habitat specialization, was also negatively affected by anthropogenic disturbances. Lastly, habitat isolation, logging and urbanization exhibited a consistent negative effect on the functional diversity of birds, whereas agriculture, habitat loss and multiple disturbances had no detectable influence. These results indicate that anthropogenic disturbances can be detrimental to birds, possibly jeopardizing their ecological functions and threatening ecosystem resilience. We therefore suggest that conservation efforts should particularly focus on reducing or mitigating the effects of these three disturbances to at least diminish their effects on avian functional diversity. Additionally, we highlight the need for novel studies to simultaneously investigate the direct relationship between disturbance, functional diversity and especially ecosystem function, through the employment of multiple complementary indices, in order to effectively assist conservation decisions.

1. Introduction

The effects of a wide array of anthropogenic disturbances (e.g. agriculture, logging, urbanization, etc.) combined with an increasing demand by human population are continuously imperiling biodiversity at global scales (Gardner et al., 2009; Laurance et al., 2014; Maxwell et al., 2016). The consequences of human pressure are not limited to species loss, affecting other biodiversity dimensions such as phylogenetic, genetic and functional diversity (Naeem et al., 2012; Mouillot et al., 2013). Specifically, functional diversity reflects the ecological trait variation of organisms (Tilman, 2001; Petchey & Gaston, 2002),

which are related to their responses to environmental changes and their effects on ecosystem functioning (Díaz & Cabido, 2001). It is commonly represented by an index that considers several trait measures simultaneously, expressing how species (and, in some cases, their abundances) are distributed in a multidimensional functional space, where the number of axes (i.e. dimensions) is determined by the number of traits and the position of species is determined by their trait values (Petchey & Gaston, 2006; Villéger et al., 2008). Additionally, the position and amount of the functional space occupied by species can be associated to their resource use in the environment (Mason et al., 2005). For instance, Petchey et al. (2004) showed that functional diversity metrics

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<https://doi.org/10.1016/j.ecolind.2020.106471>

Received 6 September 2019; Received in revised form 22 April 2020; Accepted 27 April 2020

Available online 08 May 2020

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can outperform species richness in predicting biomass production. By assuming that each species has a particular trait composition and therefore its own functional identity and ecological importance, functional diversity provides a more direct link to ecosystem functioning than the number of species (Díaz & Cabido, 2001; Mouchet et al., 2010; Cadotte et al., 2011).

Although many studies have evaluated the impacts of anthropogenic disturbances on species richness (Murphy & Romanuk, 2014; Bregman et al., 2014; Chaudhary et al., 2016), the understanding of these effects on functional diversity is still limited (Gardner et al., 2009). This occurs even for one of the best studied vertebrate groups – birds. Besides being a highly diverse group, birds play a vital role on ecosystem functioning by acting as seed dispersers, pollinators, ecosystem engineers, nutrient depositors, and insect predators (Sekercioglu, 2006; Wenny et al., 2011; Morante-Filho & Faria, 2017). Birds are greatly affected by human activities worldwide, and according to Gibson et al. (2011), especially by forest conversion into agriculture, with a higher sensitivity to habitat disturbance compared to other investigated taxa such as arthropods and mammals. Although several studies have investigated the effects of disturbances on bird functional diversity, results are greatly divergent, varying among positive (Coetzee & Chown, 2016), negative (Prescott et al., 2016) and no effects (Edwards et al., 2013). This divergence is probably due to a wide variability on the type of disturbance investigated in studies (e.g. land use change and logging; Luck et al., 2013; Chapman et al., 2018), along with the intensity or frequency (e.g. intermediate disturbance hypothesis – Biswas & Mallik, 2010), and also on the environment type (such as forests, grasslands, urban and agricultural areas). Since different types of perturbation may affect ecological traits, and consequently species' distribution in the functional space (Mouillot et al., 2013; Hatfield et al., 2018), the habitat specialization and the functional diversity index applied (e.g. functional richness, functional evenness) can also be sources of variation in the responses detected by the studies. Additionally, the functional diversity indices are directly influenced by the choice of traits included in their computation (e.g. feeding, morphological traits), which strongly varies between studies as well (Petchey & Gaston, 2006; Villéger et al., 2008), representing another possible source of variability on functional responses.

To our knowledge, no systematic review has attempted to examine the general patterns of bird functional diversity in response to different human disturbances worldwide. The available information in the literature is diffused in multiple primary studies, so a systematic approach assessing the consistency of their results can unveil general patterns and contribute to the understanding of how different human perturbations can impact on ecosystem functioning (Vandewalle et al., 2010; Bregman et al., 2014). This search for an overall effect, as well as the investigation of factors that can contribute to the variation among studies' findings, can be assessed by using a *meta*-analytical approach, which consists in a quantitative combination of individual study results into a unified general estimate (Gates, 2002; Gurevitch et al., 2018). Also, by synthesizing the existing information, this approach allows identifying priority knowledge gaps, which would be of great value for both the ecology field and the planning of conservation management strategies.

Here, we conducted the first global *meta*-analysis of the effects of anthropogenic disturbances on the functional diversity of birds to primarily investigate the existence of general patterns. In particular, we selected studies that investigated the relationship between human disturbances and functional diversity of birds through indices based on multiple traits (e.g., functional richness, evenness, divergence, and dispersion) measured in a disturbed and a control (conserved or less disturbed) treatment, or in a disturbance gradient. Additionally, we performed additional analyses to assess the potential causes of variability among studies' findings, considering the influence of study region, habitat specialization, functional traits, functional diversity index and disturbance type on species responses. We expected functional

diversity to respond negatively to disturbance; however, we predicted that the strength of functional diversity-disturbance relationships would vary according to these factors. Specifically, we expected:

(i) stronger negative relationships in tropical regions than temperate regions due to the higher sensitivity of tropical birds to perturbations (Bregman et al., 2014; La Manna & Martin, 2017), which can be explained by the lower exposition and adaptation of these species to historical disturbances such as fires, deforestation and glaciations (Betts et al., 2019);

(ii) more robust negative relationships for forest-dependent species than for all species (i.e. irrespective of habitat specialization), as the former presents specific functional traits associated to both higher habitat specialization and sensibility to disturbances (Gibson et al., 2011; Morante-Filho et al., 2015; Zurita et al., 2017);

(iii) stronger negative relationships where functional diversity indices were derived from feeding traits than non-feeding traits, given that several studies have demonstrated that certain bird trophic guilds, such as insectivores and frugivores, sharply decline in human-modified environments (e.g. Newbold et al., 2013; Burivalova et al., 2015);

(iv) functional diversity and functional richness would present the strongest negative responses to anthropogenic disturbances, given that those functional metrics are previously known to be influenced by species richness (Mouchet et al., 2010; Luck et al., 2013). In fact, several studies documented that species diversity in different taxon are negatively affected by human perturbations (e.g. Gibson et al., 2011);

(v) more pronounced negative relationships associated with habitat loss, agriculture and urbanization than other disturbances, such as patch isolation and logging. Specifically, habitat loss, agriculture and urbanization proved to be detrimental for different biodiversity dimensions of birds within local scale studies, including functional diversity (e.g. Morante-Filho et al., 2015, 2018; Prescott et al., 2016; Pena et al., 2017; Alexander et al., 2019), so these disturbances are prone to exert pronounced negative effects also at the global scale.

2. Methods

2.1. Dataset

On April 24th, 2019, we concluded our literature searches for studies that investigated the relationship between any type of anthropogenic disturbances and any functional diversity index of birds. Our searches were performed in ISI Web of Knowledge (www.webofknowledge.com) and SCOPUS (www.scopus.com) databases, restricted to the knowledge areas of interest (Agriculture, Ornithology, Biodiversity Conservation, Ecology and Biological, Multidisciplinary and Environmental Sciences), using the following search expression to filter the references: [(“functional diversity” OR “trait*”) AND (“avian” OR “bird*”) AND (“land use” OR “land-use” OR “degradation” OR “conversion” OR “alteration” OR “disturbance” OR “fragmentation” OR “perturbation” OR “habitat loss” OR “forest loss” OR “deforestation” OR “fire” OR “edge effect” OR “edge influence” OR “log*” OR “plantation*” OR “agriculture” OR “urban*” OR “dam” OR “mining” OR “clear-cutting”)]. These keywords were searched in the title, abstract and keywords of the references. In order to complement our dataset, we performed a broader search to also encompass gray literature studies, including thesis and dissertations, in Google Scholar (<https://scholar.google.com.br/>) using the expression [(“functional diversity” AND “bird*” AND (perturbation OR disturbance))].

We then performed a screening process where we eliminated duplicates and non-related themes, and selected studies that indeed compared human disturbed to control areas, as well as studies that investigated the effects of a gradient of anthropogenic perturbation on any type of multiple-trait functional diversity index of birds. We also included our own study that comprises 40 deforested landscapes (Matuoka et al., *in press*, see more details about this study in Appendix S1), and excluded articles with inappropriate or missing data. After this

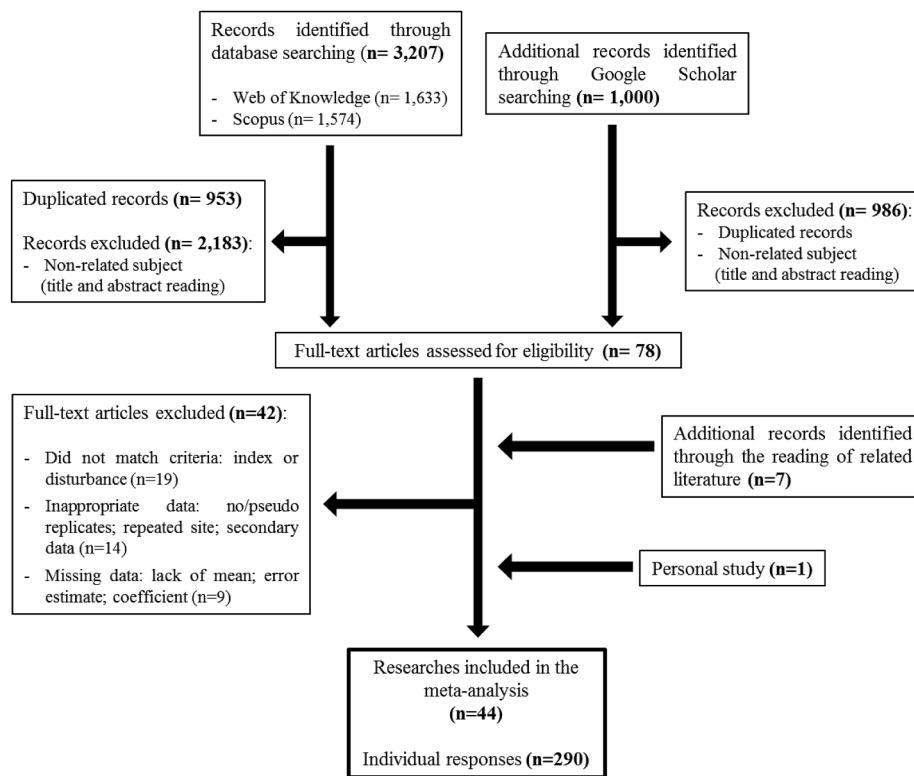


Fig. 1. Flow diagram of the dataset selection process used in our study.

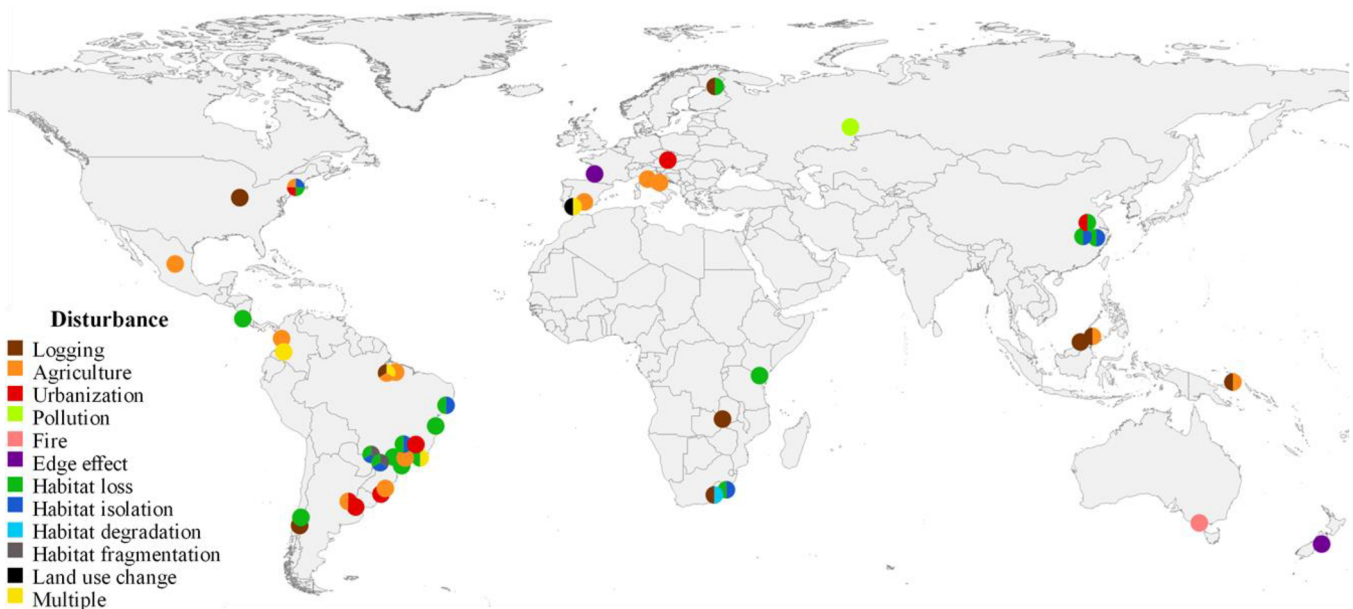


Fig. 2. Geographic distribution of studied sites included in the *meta*-analysis. Solid circles are color-coded according to the disturbance reported in each study (see legend).

screening process (Fig. 1), we ended up with 44 studies published between 2013 and 2019 (in addition to our study, which had not been published at the time of the literature searches), which were conducted in 15 countries (Fig. 2, Appendix S1). Since a single study usually employed more than one index and/or investigated the effect of more than one anthropogenic disturbance, we assembled a total of 290 comparisons (mean $\pm$ standard deviation (SD)) = 6.6 ± 5.8 comparisons per study) (Fig. 1, Appendix S1).

2.2. Data extraction and analysis

For each study that compared disturbed and control areas, we extracted the number of sites evaluated (i.e. sample size), the mean functional diversity index value and an error estimate (standard error or deviation) for each treatment. Also, we extracted the sample size and correlation coefficient (r) for studies that evaluated gradients of disturbance. When authors reported these data only in graphics, we used the DataThief (Tummers, 2006), a common program widely used in

meta-analysis studies (e.g. Arroyo-Rodríguez et al., 2017; Cattano et al., 2018) that extracts raw values (i.e. data points) directly from any digital figures (such as presented in the papers). To test our hypotheses, we also gathered information about: (i) the region where the study was performed (tropical or temperate), (ii) the habitat specialization (e.g. forest and non-forest species), (iii) trait type (feeding, morphological or various), (iv) functional diversity index (e.g. functional richness, functional evenness) employed and (v) disturbance type investigated (e.g. logging, agriculture or multiple, when more than one disturbance were considered) (see Appendices S2 and S3 for more details). We are aware that several of these disturbances are associated with each other. For instance, in human-modified landscapes, the reduction in habitat amount at the landscape scale leads to an increase in habitat isolation (Fahrig, 2003). However, we opted by separating these effects based on the categories explicitly expressed in the investigated study, to reveal their unique effects on patterns of bird functional diversity. For instance, Ding et al. (2013) assessed the effects of island area and island isolation in relation to several functional diversity indices. Their results show that island area significantly affected functional richness and evenness, whereas functional divergence patterns were better explained by island isolation. We therefore computed these results separately, i.e., included 'island area' effects into the category of 'habitat loss', and 'island isolation' into the category 'habitat isolation'.

For the mean comparisons, we first estimated the standardized mean difference between treatments (Cohen's *d*), calculated as the difference between disturbed and control treatment mean values, divided by the within-treatment standard deviations. For the correlational studies, we first converted the correlation coefficient (*r*) into the Cohen's *d*. Then, for both correlation and mean data, we applied the correction factor for small sample bias (*J*) to finally obtain the bias-corrected standardized mean difference (Hedges' *g*), which was used as the effect size measure in our meta-analysis (Borenstein et al., 2009).

We calculated the mean effect size and a 95% confidence interval for all comparisons through a random-effects meta-analysis, where negative and positive Hedges' *g* values indicate decrease and increase in functional diversity, respectively. As an attempt to correct a potential pseudo-replication bias because all studies presented more than one individual comparison, we employed a bootstrap procedure where a mean effect size was calculated for 10,000 resamples (with replacement) of only one individual comparison per study at a time (Almeida-Rocha et al., 2017). Then, a mean value and a 95% confidence interval was generated to be compared with the estimates obtained using all comparisons.

Additionally, we performed subgroup analyses to explore the heterogeneity of studies responses, particularly evaluating the influence of different regions, habitat specializations, functional traits, indices and disturbance types on the final estimates (Table 1). We did not include some comparisons employing certain indices (e.g. Functional Attribute Diversity and Mean Nearest Taxon Distance), disturbance types (e.g. fire, multiple and pollution) and habitat specializations (e.g. water-birds, non-forest and urban species) due to the small sample size (less than four comparisons by category; Table 1) that prevented us from obtaining robust estimates.

Lastly, we evaluated the relationship between the Hedges' *g* effect sizes and the predictor variables (region, habitat specialization, functional trait, index and disturbance type) through Generalized Linear Models (GLMs) using a Gaussian family. First, we removed outliers through box plot inspection, retaining 278 comparisons from the 44 studies. Then, using the Hedges' *g* value as the dependent variable, we created 32 models including all combinations of predictor variables, in addition to the null model containing only the intercept and error parameters (Table S4). Before running the models, we checked the multicollinearity between the predictor variables through the Variation Inflation Factor (VIF), and removed 14 models presenting moderate or high multicollinearity (i.e. containing variables with $VIF \geq 3$; Dormann et al., 2013). We then ran a model selection approach based on the

Akaike Information Criterion corrected for small samples (AICc) (Burnham & Anderson, 2002) and accounting for the pseudo-replication bias. We ran the models using only one comparison per study via 10,000 bootstrap procedures (with replacement) to obtain the best models (the most top-ranked within 10,000 resamples). Specifically, for each candidate model ($n = 18$), we calculated the percentage of simulations in which it was top-ranked (AICc- π_i), the mean values of AICc, Akaike weight (ω_i), and goodness-of-fit (adjusted R-squared) among all simulations. All analyses were conducted in R software (R Development Core Team, 2013), using the Metafor package (Viechtbauer, 2010).

2.3. Publication bias

Studies with non-significant results have lower chances to be published (Møller & Jennions, 2001), which can result in a biased estimation of the overall effect size in meta-analyses (Borenstein et al., 2009). To verify the robustness of the results of our meta-analysis in the face of a possible publication bias, we performed two different approaches, still considering a possible pseudo-replication bias (i.e. employing the bootstrap procedure). First, we performed the Rosenthal's fail-safe number (FSN) analysis, which calculates the number of studies with no effect that is necessary to make the overall effect size non-significant (target *P*-value = 0.05). Rosenthal's formula indicates that a result should be considered robust to publication bias if $FSN \geq 5N + 10$, where FSN is the fail-safe number obtained and *N* is the number of studies employed (Rosenthal, 1991). Second, we applied the Trim and fill method to estimate the number of missing studies that would be required to complete a symmetric funnel plot (a graphic where effect sizes are plotted against a measure of their variance, and a funnel-shaped plot indicates no biases), and then we recalculated the mean effect size including these studies (Duval & Tweedie, 2000a, 2000b).

3. Results

For all 290 comparisons obtained from 44 studies, the mean effect size was -0.14 [95% CI: -0.23 ; -0.04] and increased to -0.27 [-0.49 ; -0.05] when we used only one comparison per study, suggesting a pseudo-replication influence on the estimates. Hence, our results and discussion are based on the bootstrapped effect sizes, which indicated that disturbances have a significant negative effect on the functional diversity of birds (Fig. 3).

Regarding the publication bias, the Rosenthal's fail-safe number analysis indicated that our FSN was 1.5-fold higher than the critical value of 230, suggesting that our results are unaffected by the publication bias. The Trim and Fill analysis indicated that 0 to 12 studies could be missing, and if these studies were added our results could change from significantly negative (-0.27 [95% CI: -0.49 ; -0.05]) to no overall effect of disturbances on the functional diversity of birds (-0.14 [95% CI: -0.48 ; 0.25]). Although this test suggests that up to 12 missing studies may exist, its confidence interval includes the possibility of no missing study, which coupled with our FSN estimate and our exhaustive literature search (encompassing different databases and including gray literature), gives us confidence on our results.

The subgroup analyses revealed different patterns of bird functional responses. Specifically, we observed a consistent negative effect of disturbances on functional diversity only for studies performed in the tropical region (Fig. 3A) and that considered all species regardless of habitat specialization (Fig. 3B). There was no consistent effect among studies that investigated forest species separately (Fig. 3B). Regarding species traits, even though studies that used various traits present a marginal negative relationship with human disturbances, only studies that were based on feeding traits shared a significant negative effect (Fig. 3C). Also, the effect was consistently negative among studies that used functional diversity and functional richness indices, with the latter

Table 1

Subgroups and categories used to explore the heterogeneity of studies responses. Due to a sample size issue, we were unable to include all categories of habitat specialization, index and disturbance as presented in Table S2. Therefore, each subgroup analysis is restricted to a subset of studies and comparisons belonging to part of the categories.

Subgroup	Categories (n. of studies/n. of comparisons)
1. Region	1.1. Tropical (21/138) Latitude < 23.5° N/S 1.1. Temperate (23/152) Latitude > 23.5° N/S
2. Habitat specialization	2.1. General (35/205) All species, irrespective of habitat specialization 2.2. Forest species (8/32) Only forest-dependent species
3. Functional trait	3.1. Feeding traits (> 50% of traits belong to this category) (25/165) Traits included: diet; foraging stratum; foraging substrate; foraging method. 3.2. Morphological traits (> 50% of traits belong to this category) (5/56) Traits included: body, beak, tarsus, tail and wing measurements 3.3. Various traits (no category achieved > 50% of traits) (14/69) Traits included: feeding; morphological; breeding; habitat; life-history traits
4. Index	4.1. Functional Diversity (FD) (6/15) 4.2. Functional Dispersion (FDis) (13/38) 4.3. Functional Divergence (FDiv) (14/24) 4.4. Functional Evenness (FEve) (18/31) 4.5. Functional Richness (FRic) (24/47) 4.6. Standardized Effect Size Functional Richness (SESFRic) (6/22) 4.7. Rao's Quadratic Entropy (RaoQ) (8/20) 4.8. Mean Pairwise Distance (MPD) (4/14)
5. Disturbance	5.1. Agriculture (14/56) Studies that evaluated the effect of agricultural production systems such as: Agriculture (monoculture and mixed); Pasture; Agrosilvopastoral practices 5.2. Habitat loss (17/83) Studies that evaluated the effects of the reduction on either habitat amount in the landscape or fragment size 5.3. Habitat isolation (8/24) Studies that assessed the effects of isolation or connectivity metrics of fragments 5.4. Logging (9/50) Studies that evaluated the effects of timber harvesting processes such as: Selective logging; Partial logging; Extensive logging; Once logged; Twice logged; Partial clearcut (in patches); Secondary forest (developed after clearcut) 5.5. Urbanization (8/32) Studies that evaluated the effects of urbanization (e.g. distance to human settlements; percentage of sealed areas) 5.6. Multiple (4/13) When the study considered more than one disturbance type, but did not distinguish their effects separately

revealing the most detrimental effects (Fig. 3D). Finally, among the investigated disturbance types, only habitat isolation, logging and urbanization had a consistent negative effect on the functional diversity of birds (Fig. 3E).

The model selection approach revealed that the null model was most selected within 10,000 bootstrap resamples ($\omega = 0.51$; AICc- $\pi = 49.85\%$), followed by the region model ($\omega = 0.57$; AICc- $\pi = 45.12\%$), which explained 7–13% of the overall deviance (Table S4).

4. Discussion

Our study detected a negative overall effect of anthropogenic disturbances on the functional diversity of birds, being consistent with patterns observed for taxonomic diversity (Gibson et al., 2011; Sayer et al., 2017), and indicating that both species and functional traits are being lost in disturbed habitats. Based on this result, we investigated the possible factors responsible for the variability on the effects detected by different studies, which enabled us to understand some particularities related to bird functional responses to human disturbances.

In accordance to our prediction, we reveal that bird functional diversity is strongly and negatively affected in the tropics, but not necessarily in the temperate region. This may be explained by a higher sensitivity of tropical bird species to human disturbance, which was previously demonstrated by other studies (Bregman et al., 2014; La Manna & Martin, 2017; Betts et al., 2019). This higher sensibility may be associated with the adaptation of species to the climatic stability in the tropical region (Dynesius & Jansson, 2000; Sandel et al., 2011). In

fact, species that have evolved in environments with low levels of historical disturbances (e.g. glaciation, hurricanes, fires, storms, deforestation), such as tropical region, should be less likely to persist in the face of new disturbances, including those driven by human activities (Balmford, 1996; Betts et al., 2019). Additionally, combined with an extreme human population growth, the tropical region faces the highest deforestation and land use modification rates on Earth (Gardner et al., 2009; Laurance et al., 2014; Keenan et al., 2015).

Contrary to our expectation, we did not detect an influence of human disturbances on the functional diversity of forest-dependent species. This means that forest species present a wider range of responses to disturbances, which contradicts bird taxonomic diversity responses, since this group exhibits stronger negative responses than the complete assemblage (Gibson et al., 2011). These results can indicate that although forest-dweller species are being lost, the functional diversity of this group might be maintained or increased in human-modified habitats, which can be expected in communities exhibiting high or low degrees of functional redundancy, respectively. If species that are lost present similar traits than species that remain, functional diversity can be maintained in disturbed environments (Mayfield et al., 2010; Edwards et al., 2013). In addition, if species able to persist or colonize disturbed habitats are more functionally different from each other, a higher functional diversity in these sites can be expected (Mayfield et al., 2010; Chapman et al., 2018). This can be explained by the replacement of habitat specialist by generalist species, where the latter is more adapted to disturbances and brings different traits to the assemblage (Mayfield et al., 2010; Clavel et al., 2011). For example, we observed that within a Brazilian Atlantic forest region, forest loss can

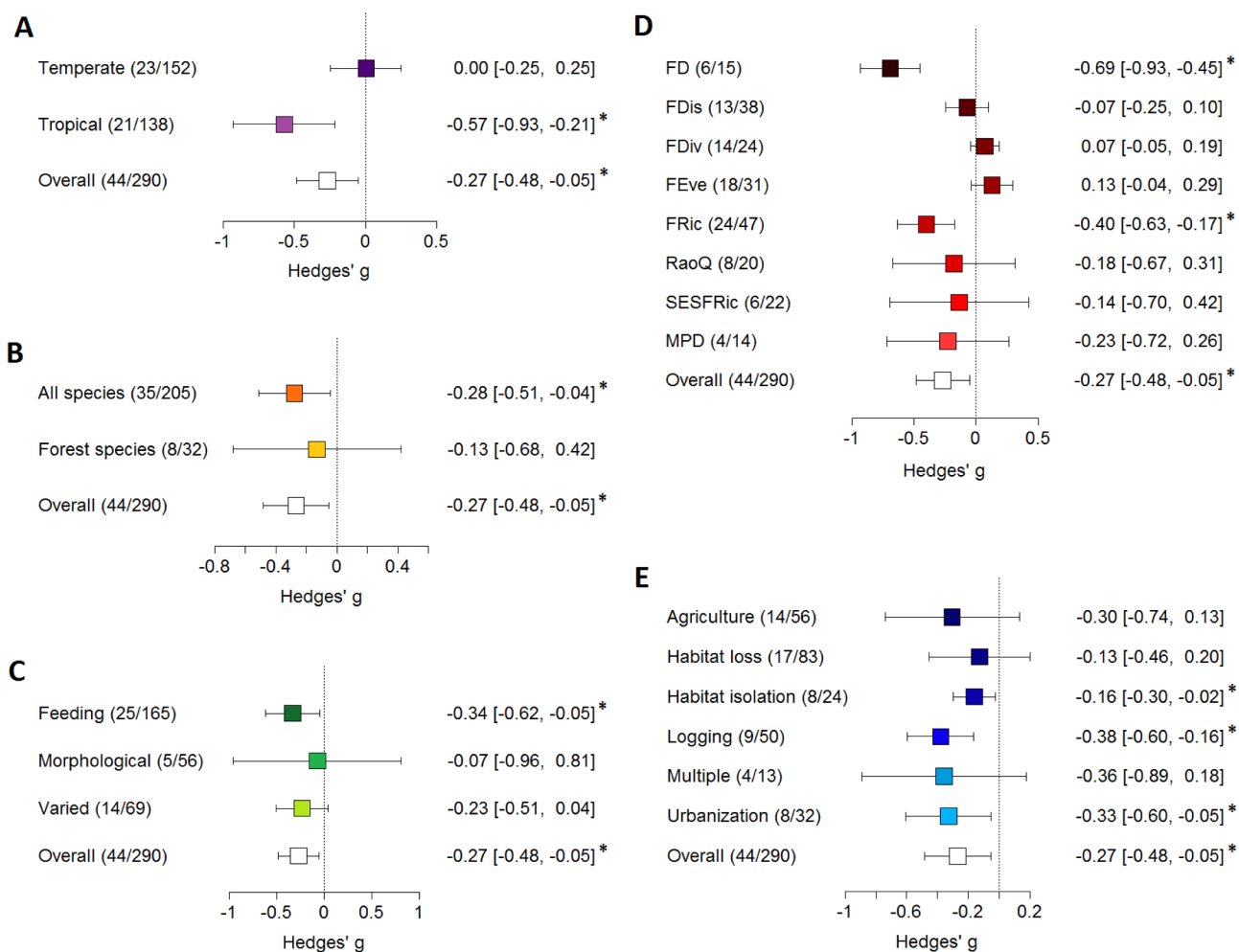


Fig. 3. Bootstrapped effect sizes of the functional diversity of birds divided by region (A); habitat specialization (B); trait group (C); functional diversity index (D) – Functional Diversity (FD), Functional Dispersion (FDis), Functional Divergence (FDiv), Functional Evenness (FEve), Functional Richness (FRic), Rao's Quadratic Entropy (RaoQ), Standardized Effect Size Functional Richness (SESFRic) and Functional Mean Pairwise Distance (MPD); and disturbance type (E). Mean effect size for all studies included (Overall). The vertical line indicates no effect of disturbances. Mean Hedges' g (squares) and the 95% CIs (lines) values are displayed on the right side (asterisks indicate statistical significance). On the left, categories of each group are presented with number of studies and individual comparisons in parentheses.

benefit birds associated with disturbed habitats, such as vultures, which presented distinct traits from the rest of the community such as scavenger diet and large body size, causing an expansion in the functional space in deforested landscapes (Matuoka et al., in press).

Another factor that adds variability among studies' findings is the type of ecological traits under investigation, with our results indicating that specifically feeding traits (i.e. diet and foraging traits, see Table 1) are negatively affected by anthropogenic disturbances. As feeding traits are directly linked to the primary function of birds (e.g. seed dispersal, pollination and insect control) (Sekercioglu, 2006), these results indicate that ecosystem functions and services performed by birds are being lost. For example, Martínez-Salinas et al. (2016) highlight the importance of birds as pest regulators, with results showing a positive association between bird trait diversity and infestation control of coffee berry borer. In contrast, we did not find any consistent pattern for studies that mostly used morphological traits (Bregman et al., 2016; Häkkinen et al., 2017; Chapman et al., 2018). Since bird species greatly vary in morphological traits such as body mass and beak size, the divergent responses can be a reflection of their different sensitivities to human disturbances (Galetti et al., 2013; Michalski & Peres, 2017).

Our findings also emphasized that the functional diversity index type also explained part of the variation among studies' findings. Currently, functional diversity can be measured by a wide variety of indices (Villéger et al., 2008), and each one captures different aspects of

this dimension. Therefore, it is plausible to expect that these indices can vary in their sensitivity to detect the effects of human disturbances. Indeed, we found that not all indices were affected by disturbances, with some indices showing great variation in their responses (e.g. Rao's Quadratic Entropy, Standardized Effect Size Functional Richness and Mean Pairwise Distance, see Fig. 3). For instance, only studies that used functional diversity and functional richness presented significant negative responses, which is in accordance to our prediction. This indicates that both the complementarity of traits and the amount of functional space occupied by species are lower in disturbed areas (Petchey & Gaston, 2002; Villéger et al., 2008). Considering that species' distribution in the functional space is associated to resource use complementarity and this can be the link between functional diversity and ecosystem functioning (Petchey & Gaston, 2002; Mason et al., 2005), low functional diversity can be related to higher niche overlap (Petchey & Gaston, 2002; Song et al., 2014), whereas low functional richness is likely associated to ineffective resources utilization (i.e. some of the potential resources remain unexploited by species; Mason et al., 2005). As these indices are prone to be positively correlated to species richness (Mouchet et al., 2010; Luck et al., 2013), they are expected to be negatively impacted in disturbed environments where species loss was already recorded.

Our results demonstrate that logging, habitat isolation and urbanization presented significant negative effects on bird functional

diversity, which corroborates with findings for amphibians (Ernst et al., 2006), trees (Nock et al., 2013) and mammals (Magioli et al., 2016). These disturbances can particularly affect some traits such as diet type and dispersal ability (Burivalova et al., 2015; Schütz & Schulze, 2015; Ehlers Smith et al., 2018), selecting species with specific traits capable to persist in disturbed environments. In fact, our results are alarming in the face of continuous urban development, habitat fragmentation and logging, especially observed in tropical regions (Gardner et al., 2009; Laurance et al., 2014). Particularly logging, currently one of the main threats to biodiversity (Maxwell et al., 2016) with enduring effects on impacted species (Burivalova et al., 2015). Additionally, when considering other traits such as conservation status and geographical range size, those rare, endemic and endangered species can be particularly prone to extinction when facing disturbances (Blackburn & Gaston, 2002; Sodhi et al., 2008; Banks-Leite et al., 2014). In this case, if these species also present unique traits and functions from the species that persist and proliferate, their local extirpation is likely to cause even worse functional decay. In contrast to our prediction, we failed in detecting an overall effect of habitat loss and agriculture on functional diversity. These results should be carefully evaluated, particularly for habitat loss, considering its direct negative effects on organisms' breeding, foraging and dispersal rates (Fahrig, 2003; Sodhi et al., 2008). Furthermore, habitat loss is likely to be associated with other disturbances herein evaluated (Laurance et al., 2014; Barlow et al., 2016). For instance, intensive logging pressure can simplify local vegetation and therefore reduce the available habitat for species (Edwards et al., 2014). Also, forest loss can lead to an increase of patch isolation at the landscape scale (Fahrig, 2003), therefore hampering species' colonization in remaining habitats.

Finally, the model selection revealed the null model as the best ranked model predicting effect size, suggesting that although the investigated predictor variables indeed add some variance among studies, none of them is strongly determinant, and the effect of human perturbation is universal (Almeida-Rocha et al., 2017). The region model was the second best-ranked model, selected with a very similar frequency than the null model. However, this model has a low explanatory power, which means that although the geographical context indeed has some influence on the response patterns, a larger part of the variance may have arisen from other reasons not explored here. For example, intensity and time since disturbance, feeding guild and geographic range of species could have important roles in determining birds' functional responses to human perturbations.

5. Conclusions

Our study revealed that anthropogenic disturbances, particularly habitat isolation, logging and urbanization, negatively affect the functional diversity of birds, especially in the tropical region. This result reinforces the urgency for effective conservation practices in tropical regions including investments in forest surveillance and restoration programs, where escalating rates of deforestation, overexploitation and urbanization have been strongly affecting biodiversity and consequently forest functioning (Gardner et al., 2009; Laurance et al., 2014).

In the face of multiple perturbations, our findings suggest that conservation efforts towards avian functional diversity should particularly focus on reducing or mitigating the effects of habitat isolation, urbanization and logging. According to studies that specifically evaluated these disturbances, a manner to lessen their impacts is through the maintenance of landscape connectivity in highly fragmented environments (Ehlers Smith et al., 2018; Anjos et al., 2019); the improvement of permeability of urban areas (e.g. increasing the proportion and diversity of native street trees; Schütz & Schulze, 2015; Pena et al., 2017); and the use of less aggressive logging practices (e.g. reduced impact logging associated to a long period of recovery post-harvesting; Edwards et al., 2013; Burivalova et al., 2015; Ibarra et al., 2017), respectively. However, due to the lack of studies, other

disturbances such as fire and pollution were left out of the analyses, despite of their relevant impact in the current scenario (Barlow et al., 2016; Mazor et al., 2018). Additionally, considering that species persistence in anthropogenic landscapes is a result of the interaction of many perturbations (e.g. deforestation, isolation and edge effect; Barlow et al., 2016), and most species are simultaneously affected by more than one threat (Maxwell et al., 2016), we stress the importance of understanding and disentangling the effects of multiple disturbances on functional diversity.

In addition, our results showed that functional diversity and functional richness indices were negatively affected by human disturbances. Although both indices are known to be influenced by species richness (Mouchet et al., 2010; Luck et al., 2013), studies revealed that they are, at least in some degree, associated to the provision of ecological functions, such as arthropod control (Philpott et al., 2009) and seed dispersal (Saavedra et al., 2014). However, studies directly investigating this relationship between functional diversity and ecosystem function are extremely scarce, as observed by Hatfield et al. (2018), who conducted a literature search and only found five studies employing this approach – two related to birds and three to plants. Thus, we emphasize that understand this relationship will contribute to unravel if functional diversity is indeed a proper mechanistic linkage between biodiversity and ecosystem functioning, and therefore can be considered as a powerful tool to help in conservation decisions.

CRediT authorship contribution statement

Maísa Assano Matuoka: Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing. **Maíra Benchimol:** Conceptualization, Methodology, Writing - review & editing. **Juliana Monteiro de Almeida-rocha:** Methodology, Writing - review & editing. **José Carlos Morante-Filho:** Conceptualization, Methodology, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the reviewers and editor for all the suggestions and improvements on the manuscript. We are grateful for all the primary studies included in our meta-analysis, in particular, we thank L. Anjos, L. Barbaro, A.L. Boesing, C.C.P. Cosset, A. Echeverri, M. Häkkinä, E. Hasui, V. Hevia, J.T. Ibarra, L.I. Jacoboski, B.T. Klingbeil, J. Leaver, C. Li, F. Morelli, F.X. Palacio, J.C.C. Pena, C. Schütz, X. Si, H. Sitters and H.G. Tripathi for data contribution. We thank the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES; 1685411/2017-2018) and the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq; 130687/2018-2019) for funding the master's scholarships to MAM. We are also grateful for the Instituto Nacional de Ciência e Tecnologia em Estudos Interdisciplinares e Transdisciplinares em Ecologia e Evolução (INCT IN-TREE) for all the insightful discussions from the workshop conducted in February 2019.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2020.106471>.

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